

DM SG 7 – Integrated Small Group (Biochemistry, Physiology, Anatomy, Histology) – Case 1 Integrated:

Dress Code: Professional Wear

How to prepare for this session:

- a) Review the lab material before coming to the lab
- b) Review the following week 6/7 lectures 11, 22, 24, 28, 30, 31, 32.
- c) Review key clinical finding in the clinical note

What to expect for the session:

1. Clinical case discussion – Case 1:

- Task 1:

Use your lectures, DLAs and textbooks to map the physiological concept/explanations of the patient's clinical findings. These clinical findings will be discussed with your peers and clinical facilitators during the small group

- Task 2:

Endeavour to correlate the laboratory findings with the signs and symptoms. Discuss the clinical and laboratory findings with your peers and clinical facilitators during the small group. Emphasis to be laid on the significance of the deranged liver function test (LFT) and histological findings. Review the basic science concepts to explain the abnormal laboratory findings.

- **At the end of the session, all students are responsible to have completed all the learning objectives listed for that session. If a group is not able to finish all learning objectives in allotted time, students have the individual responsibility for learning them. If a group finishes all learning objectives before the allotted time has ended, the students must reinforce the learned concepts by inviting interactions with facilitators, peers and subject experts.**

2. Clinical case discussion – Case 2

3. Small Group Assessment and Evaluation: Facilitator and small group evaluations

Objectives:

SOM. MK.I.BPM2.3.DM.2.BCHM.1396	Explain the biochemical mechanisms leading to steatorrhea in pancreatic disease, biliary tract obstruction, and intestinal mucosal disorders.
SOM. MK.I.BPM2.3.DM.2.BCHM.1397	Predict the digestive consequences of pancreatic insufficiency (e.g., cystic fibrosis, chronic pancreatitis), and evaluate appropriate dietary and enzyme replacement strategies.
SOM. MK.I.BPM2.3.DM.3.BCHM.1301	Explain the molecular and pathophysiologic mechanisms underlying edema, icterus (jaundice), ascites, hepatic encephalopathy, and bleeding tendency in liver disease.
SOM. MK.I.BPM2.3.DM.3.BCHM.1299	Evaluate liver function tests by comparing patterns of abnormalities in hepatocellular versus cholestatic diseases, differentiating acute and chronic liver injury, and explaining their clinical diagnostic utility.
SOM. MK.I.BPM2.3.DM.3.BCHM.1398	Identify the causes of muscle weakness and wasting in chronic liver disease, including malnutrition, electrolyte disturbances, and metabolic dysfunction.
SOM. MK.I.BPM2.3.DM.3.BCHM.1399	Identify and localize the causes of malabsorption throughout the gastrointestinal tract, from the oral cavity to the colon.
SOM. MK.I.BPM2.3.DM.3.BCHM.1400	Define dehydration, explain its causes, and describe its physiological consequences.
SOM. MK.I.BPM2.3.DM.2.BCHM.1300	Differentiate between acute and chronic liver disease using patterns in liver function test results.
SOM. MK.I.BPM2.3.DM.2.BCHM.1402	Differentiate between hepatic and posthepatic causes of jaundice based on clinical features and laboratory findings.
SOM. MK.I.BPM2.3.DM.2.BCHM.1403	Explain the metabolic basis of drug-induced liver toxicity, highlighting toxic intermediates formed during metabolism.
SOM. MK.I.BPM2.3.DM.2.BCHM.1404	Describe the biochemical basis of the clinical manifestations observed in acute alcohol intoxication, and identify factors contributing to fatty liver and complications in chronic alcoholic cirrhosis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1406	Describe the clinical features, laboratory findings and complications in chronic alcoholic cirrhosis.
SOM. MK.I.BPM2.3.DM.2.BCHM.1407	Explain the key biochemical reactions involved in nitrogen metabolism, including transamination and deamination reactions, their coenzymes, and their physiological significance.
SOM. MK.I.BPM2.3.DM.2.BCHM.1408	Describe the steps of the urea cycle, its metabolic regulation, and differentiate metabolic disorders associated with enzyme deficiencies at each step.
SOM. MK.I.BPM2.3.DM.2.BCHM.1409	Interpret laboratory findings to diagnose urea cycle disorders, create a differential diagnosis among urea cycle disorders and orotic aciduria, and analyze the metabolic fate of carbamoyl phosphate in ornithine transcarbamoylase deficiency.
SOM. MK.I.BPM2.3.DM.2.BCHM.1410	Create a differential diagnosis between all urea cycle disorders and orotic aciduria due to pyrimidine biosynthesis defect.
SOM. MK.I.BPM2.3.DM.2.BCHM.1412	Evaluate metabolic basis for treatment strategies for inborn urea cycle disorders.
SOM. MK.I.BPM2.3.DM.2.BCHM.1413	Evaluate the ethical considerations involved in genetic counseling for inborn errors of metabolism, including issues of reproductive decision-making, confidentiality, and informed consent.
SOM. MK.I.BPM2.3.DM.2.BCHM.1401	Evaluate the significance and utility of liver function tests.
SOM.MK.II.BPM2.3.DM.1.PHYS.1416	Evaluate major secretions by the exocrine pancreas with respect to their source, mechanism, composition, regulation, and functions.
SOM.MK.II.BPM2.3.DM.1.PHYS.1417	Analyze functions of the gall bladder and discuss how they are integrated with the phases of digestion.
SOM.MK.II.BPM2.3.DM.1.PHYS.1418	Compare and contrast the locations and the mechanisms that mediate the intestinal trans-epithelial movement of water, the major electrolytes, iron and calcium.
SOM.MK.II.BPM2.3.DM.1.PHYS.1419	Evaluate common causes of steatorrhea and predict effects of steatorrhea on absorption of fat-soluble vitamins.
SOM.MK.II.BPM2.3.DM.1.PHYS.1420	Differentiate secretory and osmotic diarrhea and analyze the health consequences of diarrhea like fluid loss, electrolyte disturbances, and acid-base disturbances.
SOM.MK.I.BPM2.3.DM.1.HCB.1324	Discuss the microscopic changes observed in the liver as a result of fatty liver due to alcohol liver disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1325	Discuss the microscopic changes observed in the liver as a result of cirrhosis due to alcohol liver disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1326	State the cell type involved in the formation of fibrotic tissue as a result of alcohol liver disease.
SOM.MK.I.BPM2.3.DM.1.HCB.1327	Discuss the normal microscopic structure of the cell type involved in the laying down of fibrotic tissue in the liver as a result of alcohol liver disease
SOM.MK.I.BPM2.3.DM.1.HCB.1328	Create a deeper appreciation for presentation skills and application of professionalism

Case 1:

Clinical Note:

A 55-year-old man is brought to the emergency department because of confusion for one day. His wife noted that he has been disoriented lately with decreased responsiveness. She also noted that he had repeated episodes of vomiting bright red blood that past 2 days. For the past 6 months, he complained of general loss of appetite, fatigue, unintentional weight loss, early morning nausea, and frequent abdominal discomfort. This was associated with bloating and epigastric pain that was aggravated by fatty food. He also had increased stool frequency from 1-2 per day to 6-8 per day. His stool were bulky, foul-smelling and light colored. He drinks 6 – 8 cans of beer daily for the past 30 years. He was previously admitted and treated for hypoglycemia after binge drinking alcohol for 2 consecutive days. He is currently not on any prescribed medications but he uses over-the-counter acetaminophen for the abdominal pain.

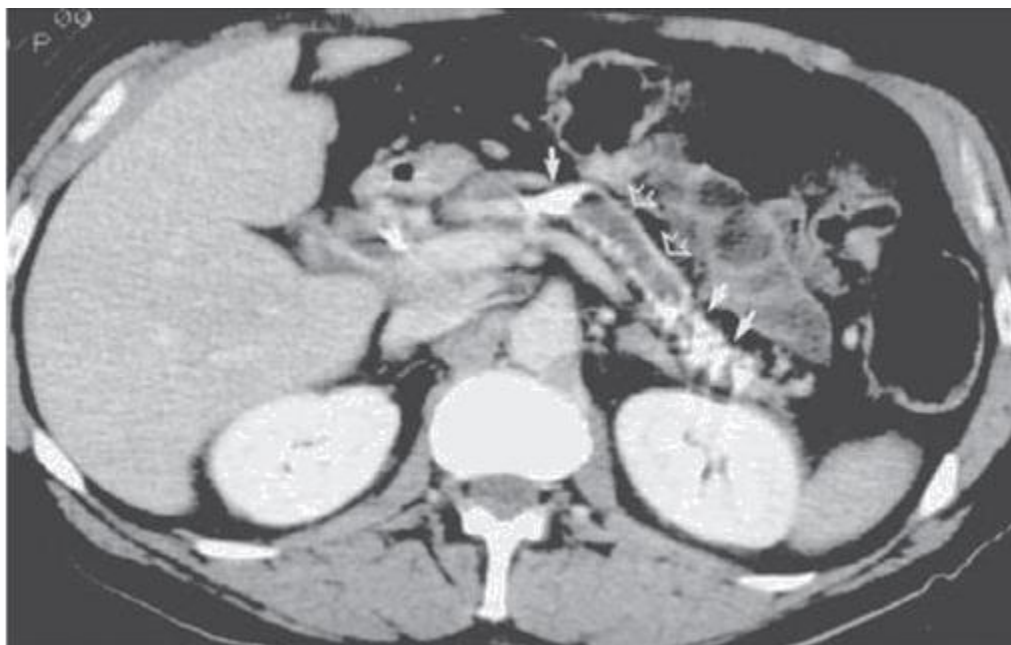
On physical examination, his blood pressure is 122/78 mmHg (supine) and 98/68 mmHg (standing), pulse 80 beats/min (supine) and 100 beats/min (standing). His skin is dry and wrinkled with reduced turgor. He has jaundice, muscle wasting, palmar erythema, and spider angioma. Abdominal examination shows several dilated superficial veins around the umbilicus, distended abdomen with shifting dullness, decreased liver span, and splenomegaly. Neurological examination shows signs of delirium, asterixis, hyperactive reflexes, and extensor toe response. Rectal examination shows internal hemorrhoids.

Lab findings show the following:

Laboratory Tests	Results	Normal Values
Three Day Stool Volume and Fecal Fat with patient on 75-100 gm/day fat		
Volume	1240 ml	< 200 ml/day
Fat	60 gm	< 6g/day
Stool microscopy	Many fat globules and some meat fibers	Very few globules and fibers
Hematocrit	43	42 – 52%
Serum cholesterol	80	80-200 mg/dL
Serum triglycerides	20	10-170 mg/dL
Plasma glucose(fasting)	220	70 – 120 mg/dL
Serum amylase	230	25 – 125 U/L
Serum lipase	415	23 – 300 IU/L
Serum potassium	2.5	3.5 – 5.0 mEq/L

Serum Calcium	6	8.4 – 10.2 mg/dL
Serum Ammonia	76 μ mol/L	15 – 45 μ mol/L
Liver Function Test:		
Total Protein (g/L)	68	60 – 78 g/L
Serum albumin (g/L)	28	35 – 55 g/L
Total bilirubin (mg/dL)	3.2	0.1 – 1 mg/dL
Direct bilirubin (mg/dL)	1.5	0.0 – 0.3 mg/dL
Alkaline phosphatase (U/L)	117	20 – 70 U/L
Alanine transaminase (U/L)	107	8–20U/L
Aspartate aminotransferase (U/L)	215	8–20U/L
γ -glutamyl transpeptidase (U/L)	512	7–45U/L
Bleeding time	More than 10 minutes	2 – 9 minutes
Platelet count (per microliters)	128,000	150,000 – 400,000
INR	1.7	1–1.1

A contrast-enhanced CT scan of the patient's abdomen shows the following radiological findings:



Task 1:

Using the SOAP note format, map the physiological concept/explanations of the patient's clinical findings.

Task 2:

Questions:

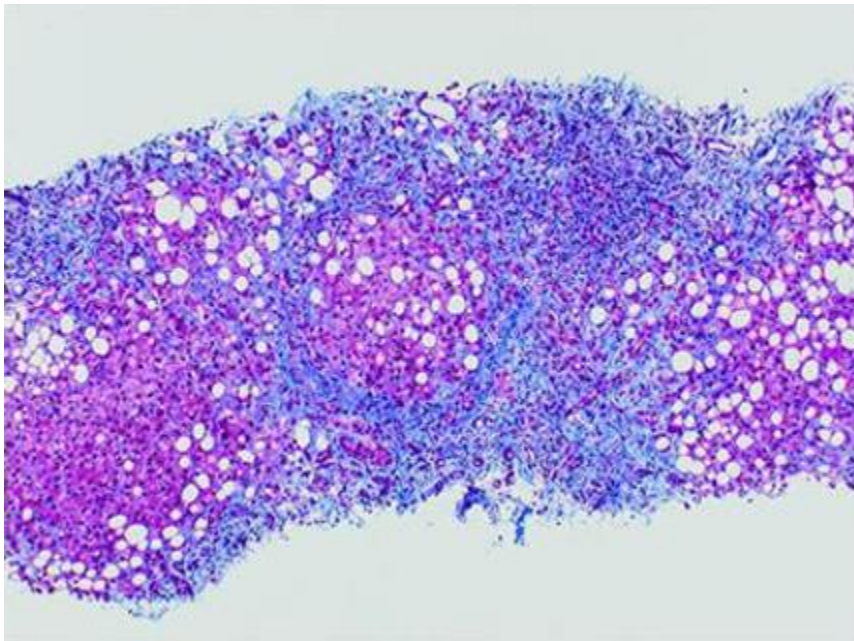
1. What is bilirubin and how is it formed?
 - How is bilirubin metabolized in the liver and indicate its further fate?
 - Indicate the reaction used to differentiate conjugated and unconjugated bilirubin in the laboratory.
2. Why is an elevated blood ammonia level toxic and which organ is most susceptible?
 - Discuss the detoxification of ammonia in the liver and the brain.
3. Why is he at increased risk of bleeding?
 - What does the elevated clotting time indicate?
 - What is the role of assessment of APTT and prothrombin time in a patient with increased clotting time?
 - Describe the vitamin required in hemostasis
4. What are the main pathways of ethanol metabolism in the liver and the principal metabolic changes that occur in chronic alcohol consumption? Discuss the differences between cytosolic versus mitochondrial aldehyde dehydrogenase and indicate which isozyme would be mainly responsible for the oxidation of acetaldehyde at 50 μM . (Their kinetic properties are summarized in Table 2)

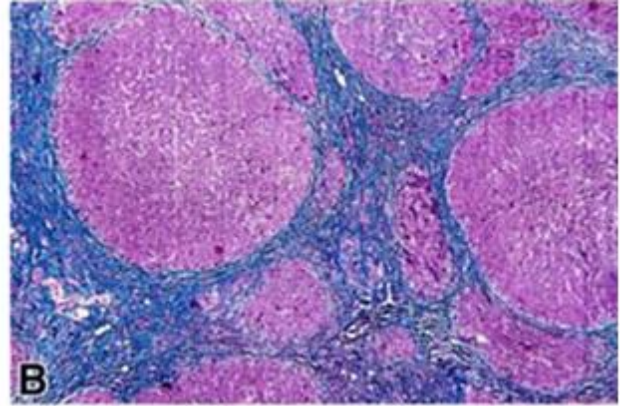
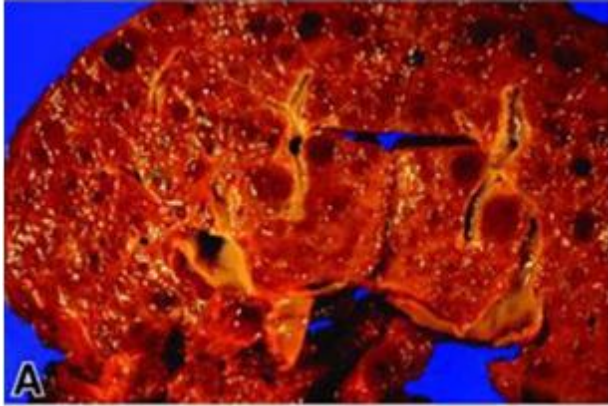
Table 2: Kinetic parameters of liver aldehyde dehydrogenases towards acetaldehyde:

Subcellular component	K _m (μM)	V _{max} (nmol/ min/ mg protein)
Cytosol	126	6.9
Mitochondrial matrix	1.4	8.1

5. What is Wernicke-Korsakoff syndrome?
 - What is the vitamin most likely deficient in such a patient?
 - Identify reactions that require this vitamin.
6. A high dose of ethanol is sometimes used as an 'antidote' to poisoning by methanol or ethylene glycol (antifreeze). What is the biochemical basis for this treatment?
7. Why should alcohol be avoided or consumed with caution when taking acetaminophen?

8. The compound disulfiram (Antabuse) has been used to prevent alcohol consumption by chronic alcoholics. How do you think that this compound would deter alcoholics from drinking alcoholic beverages?
9. Asians often exhibit extreme sensitivity to ethanol resulting in palpitations, nausea and the "flushing reaction", in which facial flushing occurs after a relatively small intake of alcohol. What is the biochemical basis for the phenomenon?
10. A person with acute alcohol intoxication is susceptible to hypoglycemia. Explain the biochemical basis.
11. In a patient with lactic acidosis, acute alcohol intoxication was suspected as the primary cause of the acidosis.
12. What are the sites of portocaval anastomosis that become enlarged in patients with portal hypertension and what are the vessels involved?
13. What histological features are typically associated with alcoholic liver disease?
 - What cells play a significant role in hepatic fibrogenesis?





14. Compare a normal CT scan of the abdomen with the patient's CT scan and identify the labeled structures below

15. What is steatorrhea? What are the possible causes of steatorrhea in this patient? Justify your answer with clinical features and laboratory findings in this patient.

16. What does a distended abdomen with shifting dullness indicate in this patient? Discuss the pathophysiologic mechanisms of ascites, spider angioma, palmar erythema, gynecomastia, asterixis, and caput medusae.

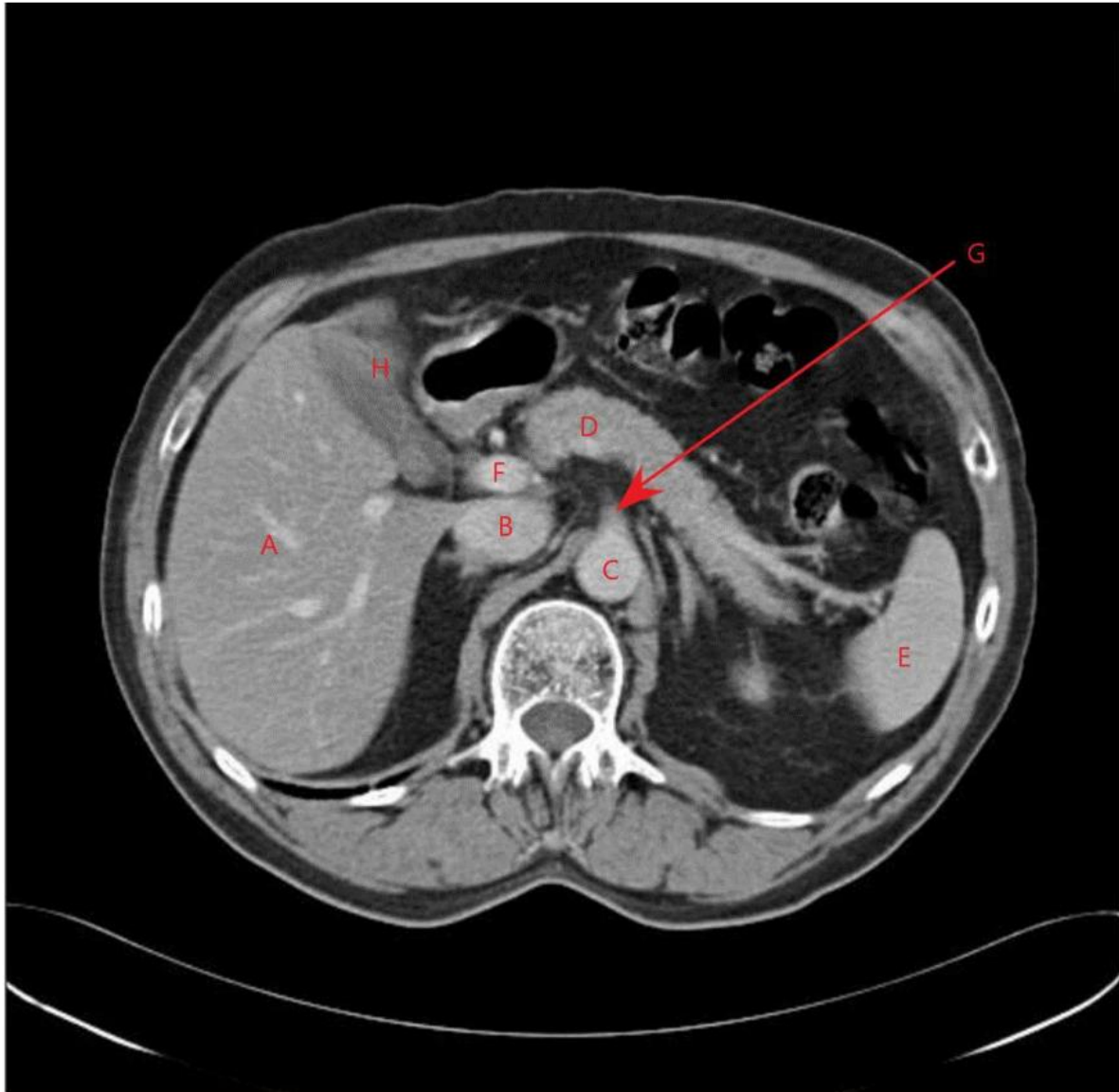


Figure: Normal CT Scan of the abdomen

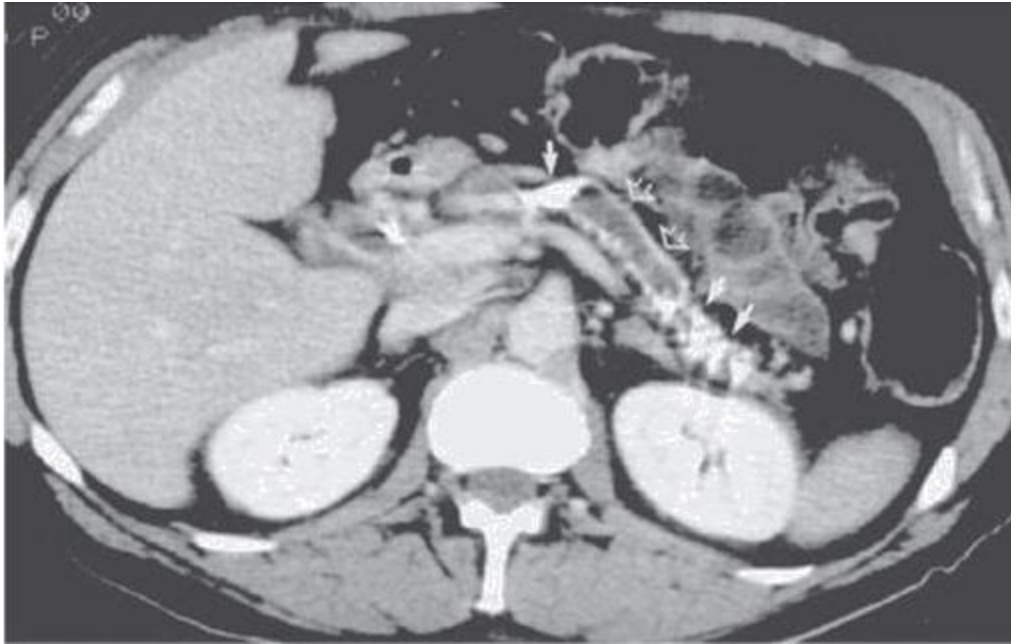


Figure: Patient's CT scan of the abdomen

Case 2 – Inborn Errors of Urea Synthesis:

Adopted and substantially modified by the SGU Department of Biochemistry, from Clinical Studies in Medical Biochemistry 3rd Edition

Edited by, Robert H. Glew & Miriam D. Rosenthal

Original case written by Pranesh Chakraborty & Michael T. Geraghty (Review the Amino acid introduction, ammonia formation / urea cycle, and specialized products of amino acids; also review pyrimidine metabolism)

Case Report:

A male infant was born at term following an uncomplicated pregnancy to a healthy woman pregnant for the first time. The delivery was uncomplicated, and growth was appropriate for the gestational age. Initially, he was breastfed every 3 hr. On the second day of life, he developed mild respiratory distress and fed poorly. Antibiotics were started, but no evidence for an infection was found. On day 3, he became increasingly unresponsive. Arterial blood gases revealed a normal P_{O_2} with a respiratory alkalosis (pH 7.56, $PCO_2 = 17$ mm Hg, $HCO_3^- 15$ mmol/L (normal 21-28 mmol/L)). Blood glucose and electrolytes were normal. He became comatose and required artificial ventilation. Repeat test showed persistent respiratory alkalosis, and the plasma ammonium ion level was 1250 μ mol/L (normal 5-30 μ mol/L). Serum electrolytes, plasma glucose, lactate, and a toxicology screen were normal. Plasma was obtained for quantitative amino acid, carnitine, and acylcarnitine analysis; urine was obtained for orotic acid measurement and organic acid profiling.

All enteral feeds were stopped. The baby was given intravenous glucose, L-arginine, sodium benzoate, and sodium phenylacetate. Hemodialysis was initiated. At this time, there were no spontaneous respirations, there was no response to painful stimuli, and brainstem reflexes were absent. The plasma amino acid results revealed a glutamine level of 1500 μ mol/L (normal 254-823), and citrulline was undetectable (normal 10 -34 μ mol/L). Quantitative carnitine, plasma acylcarnitine, and urine organic acid profiles were normal. The urine orotic acid concentration was normal at 2 μ mol/L. Over the next 12 hr, the ammonium concentration decreased to 100 μ mol/L; however, the baby remained comatose until the fifth day of life, when he began to regain brainstem reflexes and spontaneous respiration.

Blood leukocyte DNA was used for analysis of the mutated gene, and he was found to carry two separate, previously described, disease-causing mutations in the same gene, (compound heterozygote). Subsequently, the boy suffered global developmental delay, reduced head growth, and a seizure disorder. He had several hospital admissions for episodic hyperammonemia associated with intercurrent viral illnesses and died before he was three years old. The parents were advised to speak to a genetics counselor prior to planning their next child.

Questions for Inborn Errors of Urea Synthesis:

1. **Summarize the case. Recapitulate the steps in the formation of urea.**
 - What is the rate-limiting step and how is it regulated?
 - What are the various inherited urea cycle defects and associated enzyme deficiencies?
2. **Describe the importance of aminotransferase and glutamate dehydrogenase reactions in the formation of ammonia**

3. What is the biochemical enzyme deficiency in this child?
 - What is the biochemical basis of your diagnosis for this patient?
 - Give biochemical explanations for the high glutamine, low citrulline, and high blood ammonia levels observed in this patient.
4. What is the significance of estimation blood orotic acid levels in the child? If it had been elevated, what would this have signified?
 - How do you differentiate between orotic aciduria due to urea cycle defect and orotic aciduria due to defect in pyrimidine biosynthesis?
5. Carbamoyl phosphate is a substrate for two enzymes. What are the enzymes involved and describe the metabolic fates of their respective products?
6. Why was this patient given sodium benzoate, sodium phenylbutyrate and arginine as part of his treatment?
7. In addition to genetic disorders of the Urea Cycle enzymes, what other disorders present with hyperammonemia?
8. In a subsequent pregnancy the parents received genetics counseling. What is the role of the medical genetics counselor?
9. Why are the parents able to be healthy, yet they have a child who has this disorder? What is the risk that the next child born to this couple will be affected?
10. Compare some of the methods used to detect a known single base mutation in a sample of DNA obtained from a patient